

$$f(x) = x^2 \cos x$$

$$x^2 (-\sin x) + \cos(x) (2x)$$

$$h(x) = \cot(x) - e^x$$

$$\frac{\cos(x)}{\sin(x)} - e^x$$

$$\frac{\sin(x)(-\sin x) - (\cos x \cdot \cos x)}{\sin^2(x)} - e^x$$

$$\frac{-\sin^2 x - \cos^2 x}{\sin^2 x} - e^x$$

$$\frac{-1}{\sin^2 x} - e^x$$

$$19. p(t) = \cos^3(t^2 + 3t + 1)$$

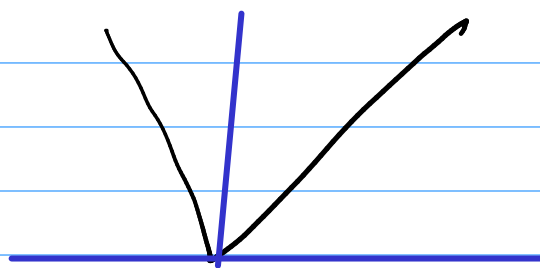
$$\left(\cos(t^2 + 3t + 1) \right)^3$$

$$3 \left(\cos(t^2 + 3t + 1) \right)^2 \left(-\sin(t^2 + 3t + 1) \right) (2t + 3)$$

$$32. g(t) = \cos(t^2 + 3t) \sin(5t - 7)$$

$$\cos(t^2 + 3t) \cdot \sin(5t - 7)$$

$$+ \sin(5t - 7) \cdot -\sin(t^2 + 3t) (2t + 3)$$



$$f(x) = 2x^3 - 9x^2$$

$$[-1, 5]$$

$$x = -1, x = 5$$

$$x = 0, x = 3$$

$$f'(x) = 6x^2 - 18x$$

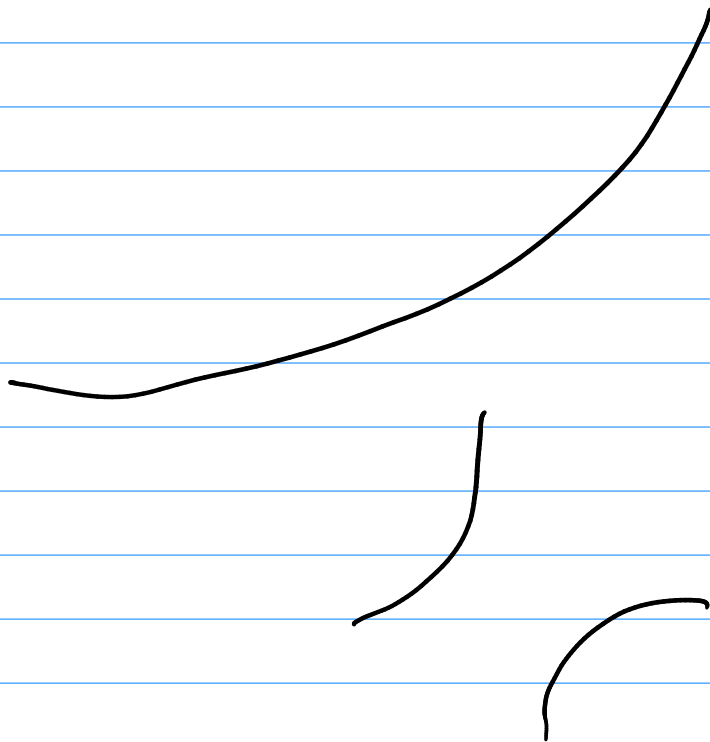
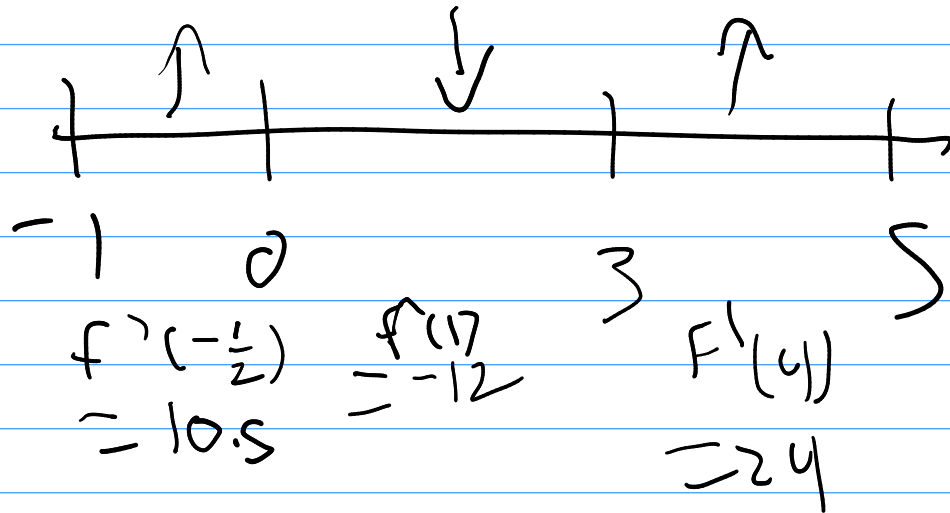
$$x(6x - 18)$$

$$f(-1) = -2 - 9 = -11$$

$$f(0) = 0$$

$$f(3) = -27$$

$$f(5) = 25$$



$$y' = 2x$$

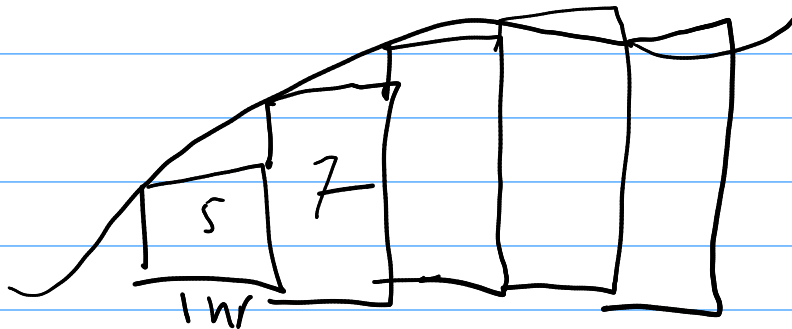
$$x^2$$

$$x^2 + \pi$$

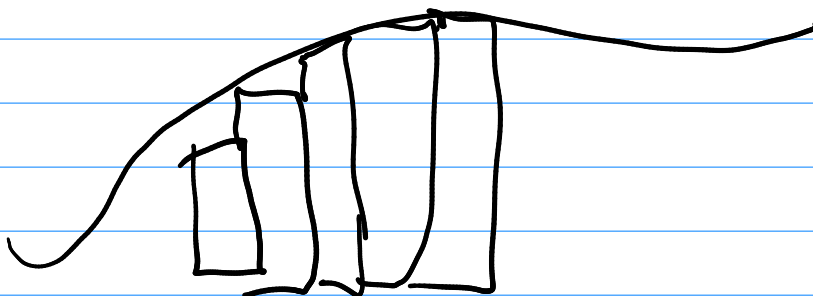
$$x^2 - 2$$

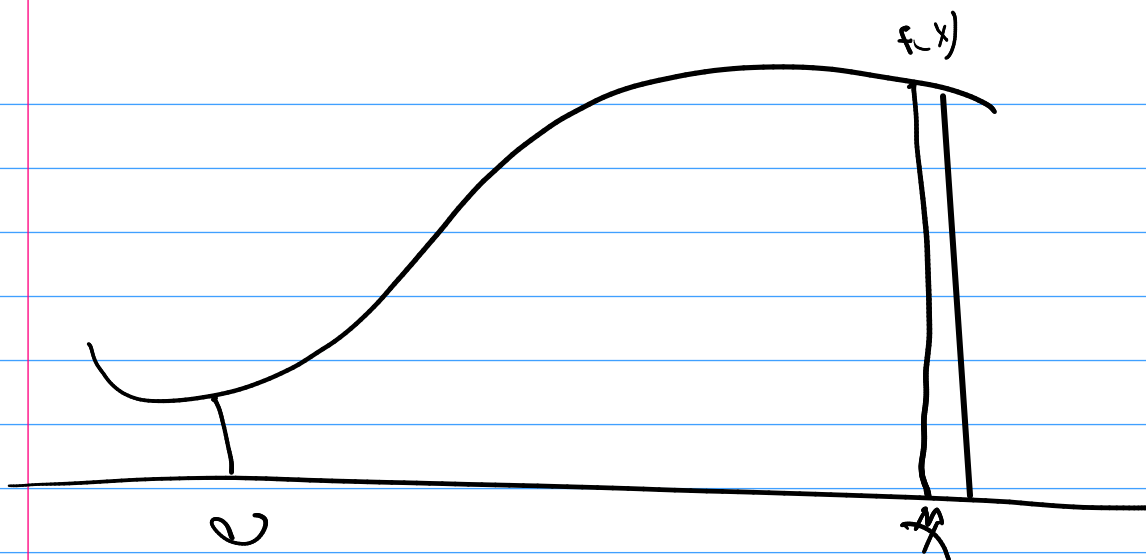
$$x^2 + C$$

$$\int 2x \, dx = x^2 + C$$



Riemann Sum





$$h \cdot f(x)$$

F T C

Fundamental
Theorem of Calculus

$$16. \int \frac{1}{\sqrt{x}} dx$$

$$\int x^{-\frac{1}{2}} dx$$

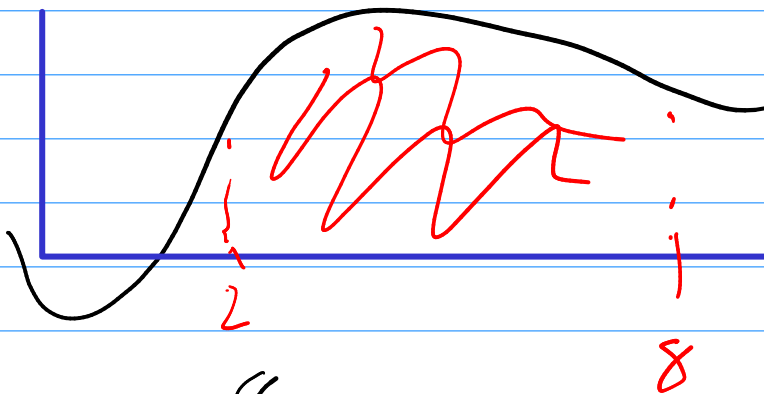
$$2x^{\frac{1}{2}} + C$$

$$x^n$$

$$\frac{x^{n+1}}{n+1}$$

$$\int (3x^2 - 6x + 1) dx$$

$$= x^3 - 3x^2 + x + C$$



$$\int_2^8 \sin x + \frac{x}{2} dx$$

$$-\cos x + \frac{1}{4} x^2 + C$$

$$\left|_2^8\right.$$

$$\left(-\cos(8) + \frac{1}{4}(8)^2\right) - \left(-\cos(2) + \frac{1}{4}(2)^2\right)$$

$$16.145 - 1.416$$

$$14.73$$

We motivate this section with an example. Let $f(x) = (x^2 + 3x - 5)^{10}$. We can compute $f'(x)$ using the Chain Rule. It is:

$$f'(x) = 10(x^2 + 3x - 5)^9 \cdot (2x + 3) = (20x + 30)(x^2 + 3x - 5)^9.$$

Now consider this: What is $\int (20x + 30)(x^2 + 3x - 5)^9 dx$? We have the answer in front of us;

$$\int (20x + 30)(x^2 + 3x - 5)^9 dx = (x^2 + 3x - 5)^{10} + C.$$

$$\int x \sin(x^2 + 5) dx.$$

u	$\frac{du}{dx}$
$x^2 + 5$	$2x$

$$\frac{1}{2} \int 2x \sin(x^2 + 5) dx$$

$$\frac{1}{2} \int \frac{du}{dx} \sin(u) dx$$

$$\frac{1}{2} \int \sin(u) du$$

$$\frac{1}{2} (-\cos u + C)$$

$$\frac{1}{2} (-\cos(x^2+5) + C)$$

$$\int \sin x \cos x \, dx$$

~~$$\int \sin u \cos u \, du$$~~

$$u = \cos x$$

$$\frac{du}{dx} = -\sin x$$

$$du = -\sin x \, dx$$

$$= \int -\sin x \cos x \, dx$$

$$= \int u \, du$$

$$-\frac{u^2}{2} + C$$

$$-\frac{\cos^2 x}{2} + C$$

$$\int \sin x \cos x \, dx$$

$$u = \sin x$$

$$du = \cos x \, dx$$

$$\int u \, du$$

$$\frac{u^2}{2} + C$$

$$\frac{\sin^2(x)}{2} + C$$

10. $\int \frac{x^3 - x}{\sqrt{x}} \, dx$

$$\int \frac{x^3 - x}{x^{\frac{1}{2}}} \, dx$$

$$\int x^{2.5} - x^{\frac{1}{2}} \, dx$$

$$\frac{x^{3.5}}{4.5} - \frac{x^{5/2}}{3/2} + C$$

$$\int \frac{x^3 - x}{x^{1/2}} dx$$

$$u = x^{1/2}$$

$$du = \frac{1}{2} x^{-1/2} dx$$

$$2 \int \frac{1}{2} \frac{x^3 - x}{x^{1/2}} dx$$

$$\frac{1}{2} \frac{1}{x^{1/2}}$$

$$2 \int u^6 - u^2 du$$

$$x^3 - x$$

$$2 \left(\frac{u^7}{7} - \frac{u^3}{3} \right) + C$$

$$2 \left(\frac{x^{7/2}}{7} - \frac{x^{3/2}}{3} \right) + C$$